Morphology-Aware Modeling of Code-Mixed Indian Languages: A Bengali-Hindi Case Study

B.Tech Project Report

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Abstract

This work presents a morphology-aware text analyzer for Bengali-Hindi code-switched speech, targeting a specific and under-studied population: Bengali L1 speakers who use Hindi as L2 in institutional settings. We collect and annotate a corpus of 42.8 minutes of naturalistic speech from 8 campus staff members at IIT Kharagpur, documenting systematic L1 transfer patterns including unidirectional masculine-default gender errors, complete ergative *ne* avoidance, and multi-level code-switching from tag-switches to full inter-sentential Bengali clauses. Our rule-based analyzer operates on Roman-transliterated text and detects gender agreement violations, ergative gaps, and code-switch boundaries with reference to curated Hindi–Bengali lexicons. A key finding is that ASR normalization of non-standard forms is **architectural, not a matter of model size or finetuning**. A four-way comparison of Whisper-large-v3, IndicWhisper-small, and IndicWhisper-large shows that all systems hide 88–100% of gender errors. Even the large Hindi-finetuned model (1.5B parameters) preserves only 2 of 16 errors, while introducing hallucination artifacts on code-switched audio. This demonstrates that morphological analysis of L1 transfer in code-switched speech requires human-corrected transcripts.

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1 Introduction

Code-switching—the alternation between two or more languages within a single conversation or utterance—is a pervasive feature of multilingual communities worldwide [Poplack, 1980, Myers-Scotton, 1997]. In India, where speakers routinely navigate between regional languages, Hindi, and English, code-switching is not a sign of linguistic deficiency but a communicative resource [Bhatia and Ritchie, 2004].

A particularly interesting case arises when speakers of a language without grammatical gender (such as Bengali) use a language that requires it (such as Hindi). Bengali marks neither gender on nouns nor gender agreement on verbs, while Hindi has a two-gender system (masculine/feminine) with obligatory agreement between nouns and verbs across tense, aspect, and number [Kachru, 2006]. When Bengali L1 speakers produce Hindi L2, they must map a genderless system onto a gendered one—a task that produces systematic and predictable errors.

This B.Tech project addresses three interconnected questions:

1. What are the systematic morphological patterns of L1 (Bengali) transfer in L2 (Hindi) speech among Bengali-dominant speakers in an institutional setting?
2. Can a rule-based text analyzer reliably detect these patterns in transcribed speech?
3. How does the choice of ASR frontend (standard Whisper vs. human correction) affect the visibility of these patterns to downstream analysis?

1.1 Motivation

Most NLP work on code-switching in Indian languages focuses on social media text [Bali et al., 2014] or targets Hindi-English mixing [Singh et al., 2018]. Bengali-Hindi code-switching in *spoken* language is comparatively under-studied, despite being common in eastern India where Bengali speakers interact with Hindi-dominant institutions. The IIT Kharagpur campus, located in West Bengal with a Hindi-medium administrative culture and Bengali-speaking staff, provides a natural laboratory for this study.

A second motivation is methodological. Off-the-shelf ASR systems like Whisper are trained on standard language and silently normalize non-standard forms. For sociolinguistic and L1-transfer research, this normalization destroys the very data of interest. We quantify this destruction and argue for human-in-the-loop transcription pipelines.

1.2 Contributions

- A curated corpus of 8 speakers (4 female, 4 male), 42.8 minutes, 3,965 words, with human-corrected Roman transliterations preserving all non-standard forms.
- A catalog of 6 morphological transfer rules (R1–R6) grounded in the data, covering gender agreement, ergative case, and code-switching.
- A rule-based `MorphologyAwareAnalyzer` implemented in Python, with curated Hindi–Bengali lexicons, that detects gender errors, ergative avoidance, and classifies code-switches.

- A four-way ASR comparison (Whisper-large-v3, IndicWhisper-small, IndicWhisper-large, and human-corrected) showing that all systems hide 88–100% of gender errors regardless of model size or Hindi finetuning. The large Hindi model (1.5B params) preserves only 2/16 errors while introducing hallucination on code-switched segments.

2 Related Work

2.1 Code-Switching in Indian Languages

Poplack [1980] established the foundational typology of code-switching: tag-switching (insertion of a discourse marker or tag from one language), intra-sentential switching (switching within a clause), and inter-sentential switching (switching at clause or sentence boundaries). This typology remains the standard framework and is adopted in our analysis.

Research on Hindi-English code-switching is extensive, driven by social media data [Bali et al., 2014, Singh et al., 2018, Khanuja et al., 2020]. The GLUECoS benchmark [Khanuja et al., 2020] provides evaluation tasks for code-mixed NLP but focuses on Hindi-English text data rather than spoken language with morphological transfer.

Bengali-Hindi code-switching has received less attention. Muysken [2000] discusses insertion patterns in South Asian language pairs, and some work exists on Bengali-English mixing [Das and Gambäck, 2014], but the specific case of Bengali L1 transfer in spoken Hindi L2 is largely unaddressed in the computational linguistics literature.

2.2 L1 Transfer and Gender Agreement

The acquisition of grammatical gender by speakers of genderless L1s is well-studied in European language pairs [Sabourin et al., 2006, Grüter et al., 2012]. The consistent finding is that speakers default to the unmarked form—typically masculine in languages like Hindi, French, and Spanish. Our data confirm this pattern: **all** gender errors in the corpus are masculine-default, with zero instances of feminine overgeneralization.

For Hindi specifically, Sharma [2004] documents gender errors in Hindi L2 speakers, and the ergative construction has been studied as a known difficulty for L2 learners [Butt, 1995]. Our contribution is the specific documentation of Bengali-to-Hindi transfer patterns with corpus evidence.

2.3 ASR for Code-Switched Speech

Whisper [Radford et al., 2023] and its multilingual variants handle many languages but are trained primarily on monolingual data. For code-switched speech, specialized systems like those from the MUCS shared task [Diwan et al., 2021] and IndicWhisper [Jain et al., 2024] attempt to handle language mixing. However, even these systems are not designed to preserve morphological errors—they aim for "correct" transcription, which in our case means hiding the phenomena under study.

3 Corpus and Data Collection

3.1 Study Design

Data was collected from 8 campus staff members at IIT Kharagpur, all Bengali L1 speakers who use Hindi as L2 in their work environment. The study was designed as a structured elicitation task with six sections:

Table 1: Questionnaire sections and target phenomena

Section	Task	Target	Elicits
A	Object naming	Vocabulary	Bengali lexical items
B	Photo comparison	Adjectives	Adjective agreement
C	Action description	Verbs	Object-verb gender agreement
D	State description	Intransitives	Subject-verb agreement
E	Free conversation	Spontaneous	Natural code-switching
F	Background	Metalinguistic	Language attitudes

3.2 Speaker Profiles

Table 2: Speaker demographics and error counts

Speaker	Gender	Role	Words	Gender Errors	Bengali Words
S1	M	Cycle shop	178*	1	15
S2	F	Staff	202	2	34
S3	F	Staff	230	2	2
S4	F	Staff	307	9	9
S5	M	Mess staff	307	0	23
S6	M	Mess staff	299	0	53
S7	M	Security	308	1	16
S8	F	Staff	229	1	7
Total			**2,060**	**16**	**159**

*S1 has two recordings (6647 + 6648); word count is the combined total.

Key observations from speaker profiles:

- Gender errors are concentrated in female speakers (S2, S3, S4, S8), consistent with R2 (masculine default in self-reference). S4 shows the highest error rate, consistent with later Hindi acquisition.
- S6 uses the most Bengali words (53) and produces full Bengali clauses, representing the high-Bengali end of the proficiency gradient.
- Male speakers (S1, S5, S7) show few gender errors because masculine default happens to be correct for male speakers' self-reference.

3.3 Transcription Pipeline

1. **Audio extraction**: WAV files extracted from interview recordings.
2. **Initial ASR**: OpenAI Whisper large-v3, language forced to Hindi.
3. **Human correction**: Each segment manually corrected against the audio, preserving all non-standard forms, hesitations, and code-switches. Bengali words transliterated to Roman script with glosses.
4. **Annotation**: Gender errors, code-switches, and Bengali insertions marked in annotator notes.

A critical finding during transcription: Whisper systematically "corrects" the data:

- Gender errors are silently fixed (e.g., speaker says *tut gaya*, Whisper outputs the "correct" *tut gayi*).
- Bengali words are converted to similar-sounding Hindi (e.g., *dorja* → *darwaza*).
- Code-switched passages are rendered as Hindi approximations.

4 Morphological Rule Catalog

We identify six systematic transfer rules from the corpus:

4.1 R1: Masculine Default on Feminine Nouns

Bengali has no grammatical gender. When producing Hindi, speakers default to the unmarked masculine verb form regardless of the noun's grammatical gender.

> **Produced:** *khursi toot gaya* (chair broke-M)
> **Expected:** *khursi toot gayi* (chair broke-F)

This rule is **unidirectional**: all errors are masculine-on-feminine. Zero instances of feminine-on-masculine were found. This asymmetry is diagnostic of L1 transfer from a genderless language, as opposed to random gender assignment.

4.2 R2: Masculine Default in Female Self-Reference

Female speakers use masculine verb forms when talking about themselves:

> **S4:** *kaam mein ghus gaya* (got-into-M work)
> **Expected:** *ghus gayi* (got-into-F)

S4 shows the highest concentration (9 detected errors), consistent with later Hindi acquisition. S2, despite being female, shows only photo-description errors—suggesting higher Hindi proficiency or metalinguistic awareness.

4.3 R3: Ergative *ne* Avoidance

Bengali has no ergative case marking. In Hindi, the ergative marker *ne* appears with transitive perfective verbs. All 8 speakers completely avoid this construction—0 instances of *ne* in 8 transitive perfective contexts detected by the analyzer.

This is an **avoidance strategy**, not an error: speakers restructure sentences to bypass the construction entirely, sometimes switching to Bengali for these contexts.

4.4 R4: Bengali Lexical Insertion (Tag-Switching)

Single Bengali content words inserted into Hindi matrix clauses, typically for household objects:

yeh toh ek toh ***dorja*** *rakhe ache* (this is a **door** kept here)

Common insertions: *dorja* (door), *janala* (window), *jol* (water), *bhat* (rice), *boi* (book), *pakha* (fan).

4.5 R5: Bengali Inter-Sentential Switching

Full Bengali clauses between Hindi utterances, most frequent in S6:

S6: *cheleti porachuna korche* (the girl is studying)
S6: *boiti ulto kore rekhechi* (kept the book upside down)

S6 sits at the high-Bengali end of the proficiency gradient; S1 and S5 also produce inter-sentential switches.

4.6 R6: Three-Way Code-Mixing

Bengali, Hindi, and English elements combined within a single phrase:

S5: *ek* ***ta*** *chair* ***eta*** (one-CLF chair here)

Bengali classifier *ta* + English *chair* + Bengali deictic *eta*.

5 System Architecture

5.1 Pipeline Overview

```
Audio (.wav)
  |
  v
ASR Frontend (Whisper / IndicWhisper)
  |
  v
[Human Correction] (corrected_text in Roman transliteration)
  |
  v
MorphologyAwareAnalyzer
```

```
  |-- Word-level Language ID
  |-- Gender Error Detection (R1, R2)
  |-- Ergative 'ne' Analysis (R3)
  |-- Code-Switch Classification (R4, R5, R6)
  |
  v
Linguistic Annotations (per-segment JSON)
```

5.2 Lexicon Design

The analyzer relies on curated lexicons in Roman transliteration:

- **Hindi feminine nouns** (60+ entries): *kursi, kitaab, roti, gaadi, chaabi, khidki, …*
- **Hindi masculine nouns** (30+ entries): *darwaza, pankha, ghar, paani, …*
- **Bengali words** (55+ entries with Hindi equivalents): *dorja→darwaza*, *janala→khidki*, …
- **Verb patterns** (80+ masculine, 80+ feminine): Perfective (*gaya/gayi*), habitual (*karta/karti*), progressive (*raha/rahi*)
- **English common words** (150+ entries): For language identification

5.3 Gender Error Detection

The algorithm operates in two modes:

R1 (noun-verb): For each feminine noun in the text, scan a 5-word window for compound masculine verb forms. Only compound patterns (*tut gaya*, *khol raha*) are matched to avoid false positives from single-word forms that could agree with an implicit subject. Longest patterns are matched first.

R2 (subject-verb): For female speakers only. Checks for masculine verb forms in self-referential contexts: first-person pronouns (*mai, hum, mera*) within a 6-word window, or habitual/past-habitual forms without a third-person subject. Photo-description contexts are excluded.

5.4 Code-Switch Classification

1. Words are classified as Hindi, Bengali, English, or unknown via lexicon lookup and morphological suffix matching.
2. Bengali word positions are grouped into contiguous runs (allowing gaps of up to 2 words for particles/determiners).
3. Runs are classified:
   - 1 word = **tag-switch** (R4)
   - 2–3 words = **intra-sentential** (R4)
   - 4+ words = **inter-sentential** (R5)
   - Bengali + English nearby = **three-way** (R6)

5.5 Devanagari Transliteration

Whisper outputs Devanagari script. For ASR comparison, a basic Devanagari-to-Roman transliterator maps characters to their informal Roman equivalents (e.g., क→*k*, ख→*kha*), with schwa deletion at word boundaries. This is approximate but sufficient for lexicon matching.

6 Evaluation

6.1 Analyzer Performance on Corrected Transcripts

Running the analyzer on human-corrected transcripts:

Table 3: Analyzer output on corrected transcripts

Metric	**Count**
Total words analyzed	2,060
Language identification	
Hindi	67.2%
English	12.7%
Unknown	12.4%
Bengali	7.7%
Gender errors detected	16
Noun-verb (R1)	–
Subject-verb (R2)	–
Code-switches detected	78
Transitive perfective contexts	8
With *ne*	0

The "unknown" language category (12.4%) consists primarily of proper nouns (Kharagpur, IIT), numbers, hesitation markers (*uhhh*), and bracket annotations ([unclear]). The lexicon covers 87.6% of tokens.

6.2 ASR Comparison: Corrected vs. Whisper vs. IndicWhisper

We compare four text sources: human-corrected transcripts (gold standard), Whisper-large-v3 raw output, and two IndicWhisper variants—`vasista22/whisper-hindi-small` (244M parameters) and `vasista22/whisper-hindi-large-v2` (1.5B parameters), both Hindi-finetuned Whisper models.

Table 4: Four-way ASR comparison: impact on detected phenomena

Phenomenon	**Corrected**	**Whisper-v3**	**Indic-small**	**Indic-large**
Total words	2,060	2,957	742	2,209
Gender errors	16	0	0	2
Code-switches	78	48	8	40
Bengali words	159	52	9	45
Hallucination segs.	—	—	2	6
Percentage of phenomena hidden (vs. corrected):				
Gender errors	—	100%	100%	88%
Code-switches	—	38%	90%	49%
Bengali words	—	67%	94%	72%

Key findings:

1. **Gender error normalization is near-total across all models.** Whisper-large-v3 and IndicWhisper-small hide 100% of gender errors; IndicWhisper-large preserves only 2 of 16 (88% hidden). The two surviving errors are in S4 and S8, suggesting that the large model occasionally reproduces non-standard forms when the speaker's actual pronunciation is sufficiently clear, but this is unreliable.

2. **Model size matters for coverage but not for morphological fidelity.** IndicWhisper-large (2,209 words) achieves comparable coverage to Whisper-large-v3 (2,957 words), while IndicWhisper-small produces only 742 words—a 64% loss. However, even with adequate word counts, the large Hindi model still hides 88% of gender errors.

3. **Bengali word recognition improves with scale but remains poor.** IndicWhisper-large detects 45 Bengali words (72% loss) vs. IndicWhisper-small's 9 (94% loss). Whisper-large-v3, being multilingual, detects 52 (67% loss). Hindi-finetuned models convert Bengali words to Hindi-sounding equivalents more aggressively than the multilingual baseline.

4. **Hallucination increases with Hindi finetuning.** IndicWhisper-large produces 6 hallucination segments (repetition loops like "हाउस" ×60 and fabricated news-article text), while the small model produces 2 and Whisper-large-v3 produces none. Code-switched audio appears to trigger the hallucination failure mode in Hindi-finetuned models, which expect monolingual input.

5. **Whisper-large-v3 remains the best ASR baseline overall**, with the highest code-switch preservation (48/78) and Bengali word detection (52/159), and no hallucinations. However, its 100% gender error normalization means it still cannot serve as a transparent frontend for morphological analysis.

These results demonstrate that the ASR normalization problem is **architectural, not a matter of model size or language-specific finetuning**. All four configurations—multilingual large, Hindi-finetuned small, and Hindi-finetuned large—hide the majority of non-standard morphological forms. Human-corrected transcripts are necessary, not merely preferable, for studying L1 transfer in code-switched speech.

6.3 Section-Level Error Distribution

Table 5 breaks down detected phenomena by questionnaire section. This reveals where each transfer pattern is most active.

Table 5: Phenomena by questionnaire section

Sec.	Task	Segs	Words	Gender Err.	Switches	Bengali
A	Naming	18	270	2	11	36
B	Adjective	5	74	2	4	4
C	Action	11	206	3	16	35
E	Conversation	44	851	5	29	37
F	Background	42	659	4	18	47
	Total	**120**	**2,060**	**16**	**78**	**159**

Key observations:

- Section A (naming) elicits the most Bengali lexical items per word, confirming that household vocabulary triggers L1 access.
- Section C (action descriptions) has the highest code-switch density (16 switches in 206 words = 7.8%), because speakers describe photos using Bengali verb forms.
- Gender errors are distributed across all sections but are most frequent in free conversation (E), where speakers produce longer, less monitored speech.
- Section D (intransitive states) is subsumed into sections A and C in practice, as speakers describe broken/open/fallen objects within the same photo-description task.

7 Discussion

7.1 Unidirectionality of Gender Errors

The most striking finding is the **complete unidirectionality** of gender errors. All errors are masculine-default; no speaker ever produces a feminine verb with a masculine noun. This pattern is predicted by the Transfer Hypothesis: Bengali speakers, lacking gender categories in their L1, default to the unmarked (masculine) form. Random assignment would produce errors in both directions.

7.2 Proficiency Gradient

Error frequency correlates inversely with Hindi exposure:

- S4 (9 errors): Later Hindi acquisition, limited exposure.
- S2 (2 errors): Longer Hindi exposure, only photo-description errors.
- S5, S6 (0 errors): Male speakers—masculine default is correct for self-reference.

7.3 Avoidance as Strategy

The complete absence of ergative *ne* (0/8 transitive perfective contexts) is an avoidance strategy, not ignorance. Speakers restructure sentences or switch to Bengali rather than producing incorrect *ne* constructions. This is linguistically more sophisticated than error production and consistent with the Avoidance Hypothesis [Schachter, 1974].

7.4 Limitations

1. **Corpus size**: 8 speakers and 2,060 words is small for statistical generalization, though sufficient for documenting qualitative patterns.

2. **Rule-based approach**: The analyzer cannot detect novel patterns not encoded in its lexicons and rules. A machine learning approach would require a larger annotated corpus.

3. **Unknown words**: 12.4% of words are classified as "unknown," primarily proper nouns and numbers outside the lexicon's scope.

4. **Gender error precision and recall**: An initial run of the analyzer flagged 36 candidate errors. We manually audited every one against its full segment context and found 17 false positives (wrong noun gender in the lexicon, third-person or impersonal subjects, mixed-gender group subjects, and duplicate R1/R2 detections). After fixing these rules, the analyzer reports 16 errors, all of which are confirmed by the audit (precision 16/16 on the audited set). Two audited true positives (S4, "*kaam karta tha*" and "*ghus gaya*" in a narrative with no first-person marker) are missed, because identifying the speaker as the subject requires discourse-level coreference that the rule-based checker does not perform (recall 16/18). Because the rules were tuned on this same audit, these figures are optimistic and should be validated on held-out speakers.

8 Conclusion and Future Work

We have presented a morphology-aware analyzer for Bengali-Hindi code-switched speech that detects L1 transfer patterns invisible to standard ASR. The system demonstrates that:

1. Gender agreement errors in L2 Hindi are **systematic, unidirectional, and predictable** from the L1 (Bengali) typology.

2. **ASR normalization is architectural, not fixable by finetuning.** A four-way comparison (Whisper-large-v3, IndicWhisper-small, IndicWhisper-large) shows all systems hide 88–100% of gender errors. Hindi finetuning introduces hallucination on code-switched audio without preserving non-standard morphology.

3. A rule-based analyzer with curated lexicons can detect these patterns with reasonable accuracy on corrected transcripts.

8.1 Future Directions

- **Larger corpus**: Expanding to more speakers and longer recordings for statistical analysis.
- **Other L1-Hindi pairs**: The framework is extensible to other genderless L1s (e.g., Tamil, Telugu) transferring into Hindi.
- **Neural approach**: Training a sequence labeling model on the annotated data for joint language ID and error detection.
- **Real-time feedback**: Integrating the analyzer into a language learning tool for Bengali speakers learning Hindi.

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